

SAMUEL MARTINEZ

Full Stack & AI Engineer

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PROFESSIONAL SUMMARY

Full Stack and AI Engineer with 5+ years shipping production software and nearly 2 years building AI systems used by real customers at scale — multi-agent orchestration, LLM-powered backends, and production-grade TypeScript/Node.js and Python/FastAPI services. Track record of taking generative AI from prototype to high-availability, observable production systems handling hundreds of thousands of monthly requests.

EXPERIENCE

Lead AI Engineer — Elaniin

Dec 2024 – Present

Remote

SpecAgent — AI QA Platform for Government Digital-Procedures System (El Salvador)

Core — multi-agent backend • Python 3.12 • FastAPI • Google ADK • Claude (Vertex AI) • PostgreSQL 18

- Designed and built a multi-agent AI backend (Google ADK + Claude/Vertex AI) with 4 specialized agents that interview users to author 6-section QA specifications, generate test cases from approved specs, review and critique output, and build QA artifacts — reducing manual spec-writing effort for QA teams
- Architected a ~25K-LOC FastAPI service with async SQLAlchemy 2.0/PostgreSQL 18, Alembic migrations, Keycloak/JWT auth (python-jose), and real-time SSE updates, deployed to Google Cloud Run across 4 environments via GitLab CI (lint, test, SCA, SonarQube) with pytest/ruff/pre-commit quality gates
- Integrated the Qase test-management API for bidirectional sync of AI-generated test cases and live test-run results pulled back into the platform

Automation — AI-generated E2E framework • TypeScript • Playwright • custom Claude Code skill

- Built an AI-driven test-generation pipeline: authored a custom Claude Code skill (Plan→Build) that consumes approved QA specs via the Core API and generates Playwright/TypeScript suites — page objects, fixtures, and assertions — with zero manual test authoring
- Designed a role-based Page-Object-Model framework (citizen, analyst, inspector, and lead personas) with reusable form/assertion helpers for testing complex, stateful government workflows, including flaky-UI handling and Qase-linked reporting
- Codified generation conventions (selector priority, naming, file structure) enforced automatically by the AI generation skill to guarantee consistency across auto-generated test code

Web — review & monitoring console • Next.js 16 (App Router) • React 19 • Tailwind CSS 4

- Built a ~21K-LOC Next.js 16/React 19 frontend (Atomic Design structure) featuring a real-time AI chat interface (SSE streaming with tool-call visualization) for collaborative spec authoring, a TipTap Markdown editor, and NextAuth v5 + Keycloak authentication
- Implemented a master-detail test-case review workflow with AI-assisted generation, inline editing/diffing, one-click Qase sync, and live pass/fail status reporting
- Set up authenticated SSE streaming over fetch/ReadableStream (bypassing native EventSource's header limitations) and enforced 60%+ test coverage via Vitest/Testing Library with Playwright E2E for complex stateful views

Selected AI Projects

- AI Customer Service/Sales Agent** — deployed at scale, handling hundreds of thousands of monthly requests; drove record client sales and satisfaction gains
- Document AI/OCR pipelines (2 clients)** — TypeScript/GCP JSON extraction from legal docs and government IDs w/ fraud-detection logic; cut processing time 60%
- Real-time Concierge Agent (Gemini Live API/WebSockets) & Proposal-Drafting Agent (advertising)** — tool-calling agents for live event registration and data-driven proposal generation

Full Stack Engineer — Elaniin

Apr 2023 – Dec 2024

Remote

- Automated visa/residency application processing for the Salvadoran government, cutting response times for reports, fines, and extensions by 80%
- Refactored code into reusable components/design patterns and enforced TDD/DDD methodologies to improve reliability and maintainability

Full Stack Engineer — ProGPS

Jul 2022 – Apr 2023

Remote

- Delivered custom React.js/Laravel solutions for client business requirements; optimized Laravel queries and performance; wrote testable, SOLID-compliant code

- Migrated legacy Vue.js 2 components to Vue.js 3/TypeScript; built a Python socket-based desktop app connecting Jackpot machines to computers; developed high-performance Laravel API endpoints

PERSONAL PROJECTS

- **AI Lead Generator (Telegram-controlled multi-agent system)** — TypeScript/Node 22 orchestrator coordinating Discovery, Qualification, and Outreach agents (Google Gemini) to find, score, and draft outreach for leads; RAG via PostgreSQL/pgvector, MCP tools (Playwright, Google Search), Drizzle ORM, Langfuse/OpenTelemetry observability, Sentry error monitoring
- **haku-ai.com** — AI resume & job-application platform: Node.js/Express backend with JWT auth, PostgreSQL/Knex, resume parsing (PDF/DOCX), and GPT-4-powered resume analysis, ATS optimization, and cover-letter generation

EDUCATION

Engineering in Computer Science — Universidad Tecnológica de Santiago (UTESA)

2019 – 2025

SKILLS

Languages: TypeScript, JavaScript, Python, PHP, SQL

Backend & Frameworks: Node.js, NestJS, Express.js, FastAPI, Next.js, React, Laravel

Data & Infra: PostgreSQL, SQLAlchemy (async), Alembic, Drizzle ORM, Prisma ORM, Docker, AWS, GCP (Cloud Run), Digital Ocean, GitLab CI/CD, Keycloak/JWT, SSE

Generative AI: Claude API (Vertex), Claude Code skills, Gemini API, Gemini Live API, OpenAI API, Google ADK, RAG, LangChain, Vertex AI, prompt engineering, multi-agent orchestration, tool calling

Testing & Observability: Vitest, Testing Library, Playwright E2E, pytest, Qase, Langfuse, OpenTelemetry, ruff/ESLint/pre-commit